

AWS DevOps Engineer Intern Assignment Report

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Objective

Deploy a simple website on an AWS EC2 instance using Nginx and document the complete process.

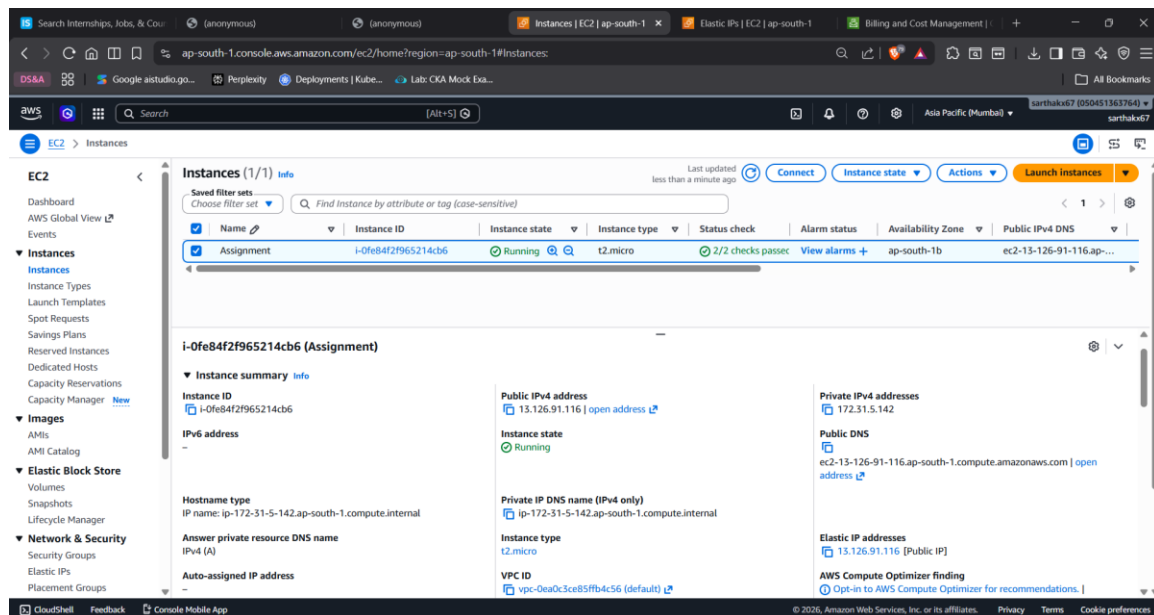
Services Used

- EC2
- Security Groups
- VPC (Default)
- Internet Gateway
- Elastic IP
- Docker in EC2
- Shell Script

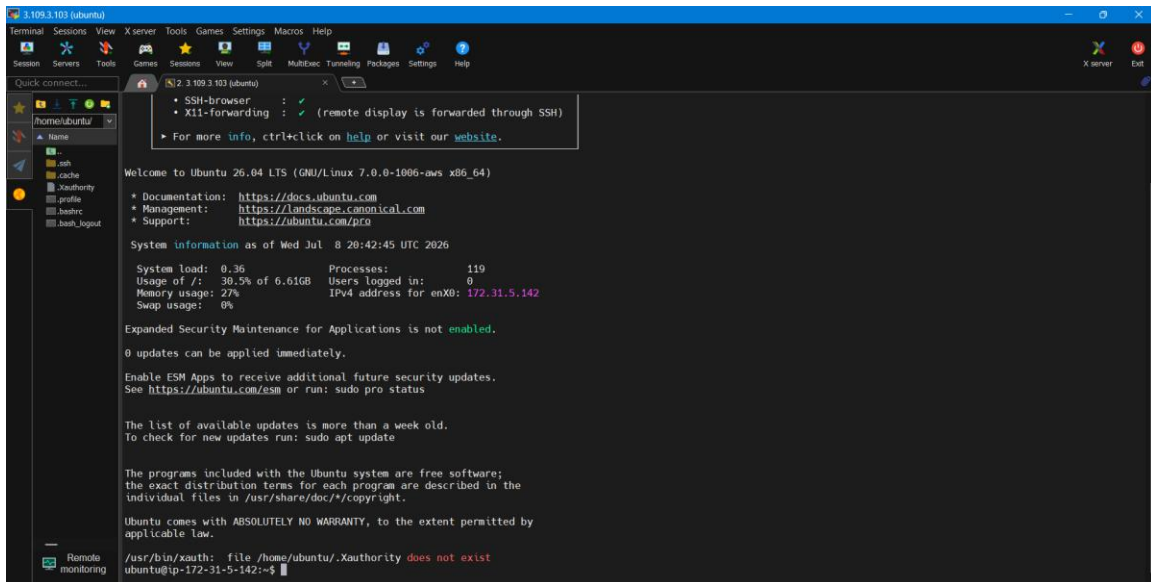
Task 1: EC2 Instance Setup

Describe the instance configuration.

1. created aws instance



2. ssh into aws instance

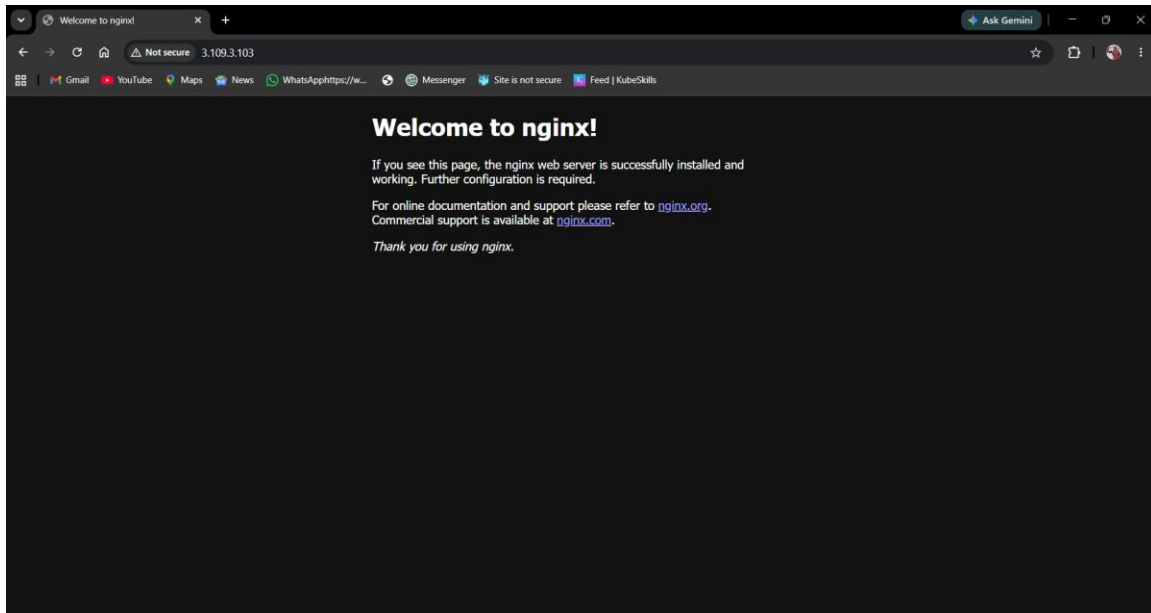


3. Installed nginx

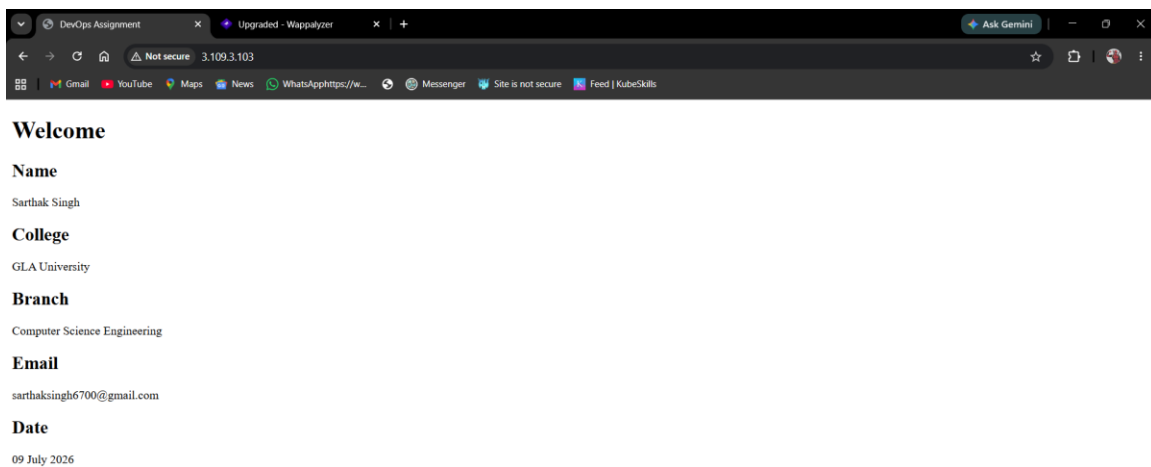
Commands Executed:

- sudo apt update
- sudo apt install nginx -y
- sudo systemctl status nginx
- sudo systemctl restart nginx
- df -h
- free -h
- ps aux

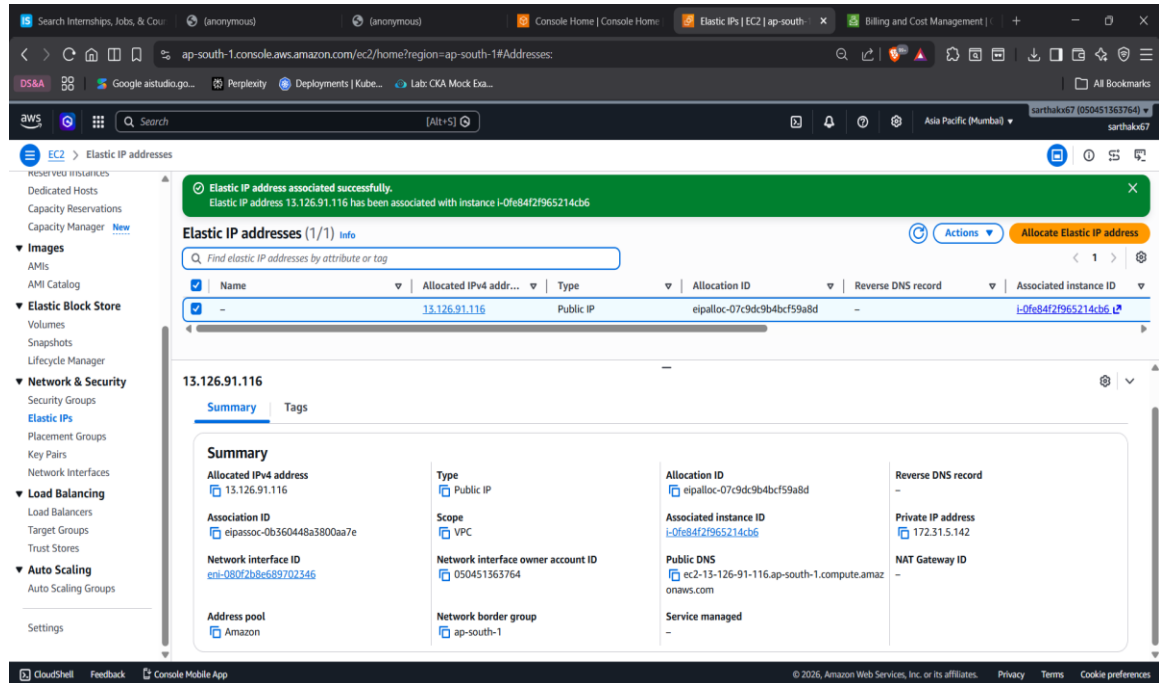
Default Nginx Page



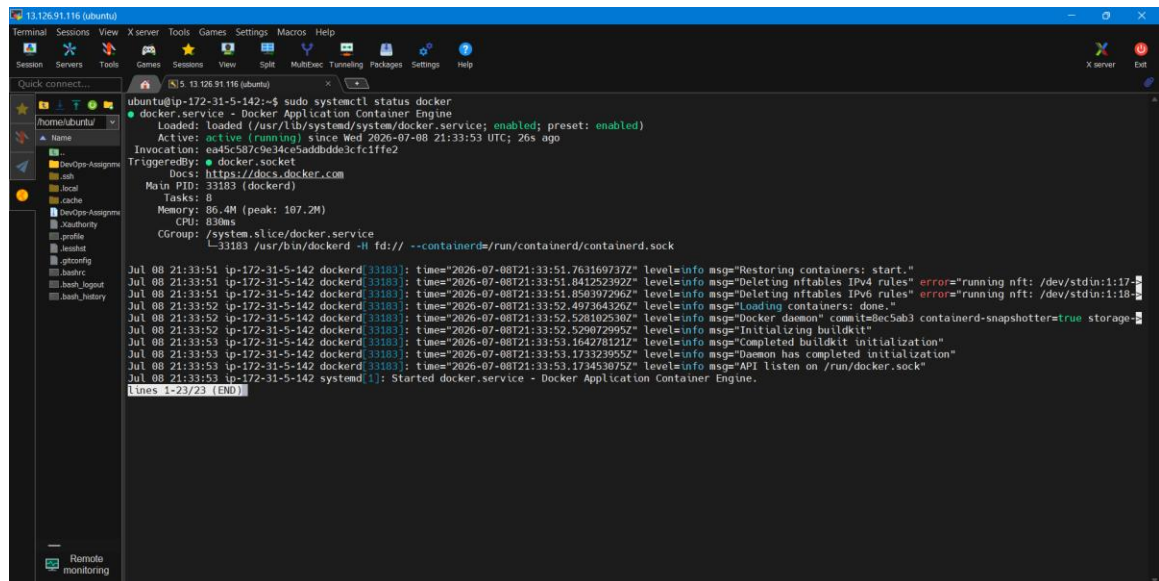
4. Changed index.html File content



3. Added Elastic ip to aws Instance



4. Installed Docker and run Hello-World Docker Container



```
Terminal Sessions View X server Tools Settings Help
Session Servers Tasks Games Stars View Split Multitask Tunneling Packages Settings Help

Quick connect...
home/ubuntu@ ~
$ sudo docker run hello-world
Jul 08 21:33:51 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:51.850397266Z" level=info msg="Deleting nftables IP6 rules" error="running nft: /dev/stdin:1:10:"
Jul 08 21:33:52 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:52.407364267Z" level=info msg="Loading containers: done."
Jul 08 21:33:52 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:52.528102530Z" level=info msg="Docker daemon" commit=8ec5ab3 containerd-snapshotter=true storage=
Jul 08 21:33:52 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:52.529872995Z" level=info msg="Initializing buildkit"
Jul 08 21:33:53 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:53.164278121Z" level=info msg="Completed buildkit initialization"
Jul 08 21:33:53 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:53.173323955Z" level=info msg="Daemon has completed initialization"
Jul 08 21:33:53 ip-172-31-5-142 dockerd[31813]: t=me="2026-07-08T21:33:53.173453875Z" level=info msg="API listen on /run/docker.sock"
Jul 08 21:33:53 ip-172-31-5-142 systemd[1]: Started docker.service - Docker Application Container Engine.

ubuntu@ip-172-31-5-142:~$ "[[200-sudo docker run hello-world->C
ubuntu@ip-172-31-5-142:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
Latest: Pull from library/hello-world
4f50b58fddms: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:96498ffdc522e78bb7ab6384a5c0485a79b9c7c08ca79ba08623edcad105462d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

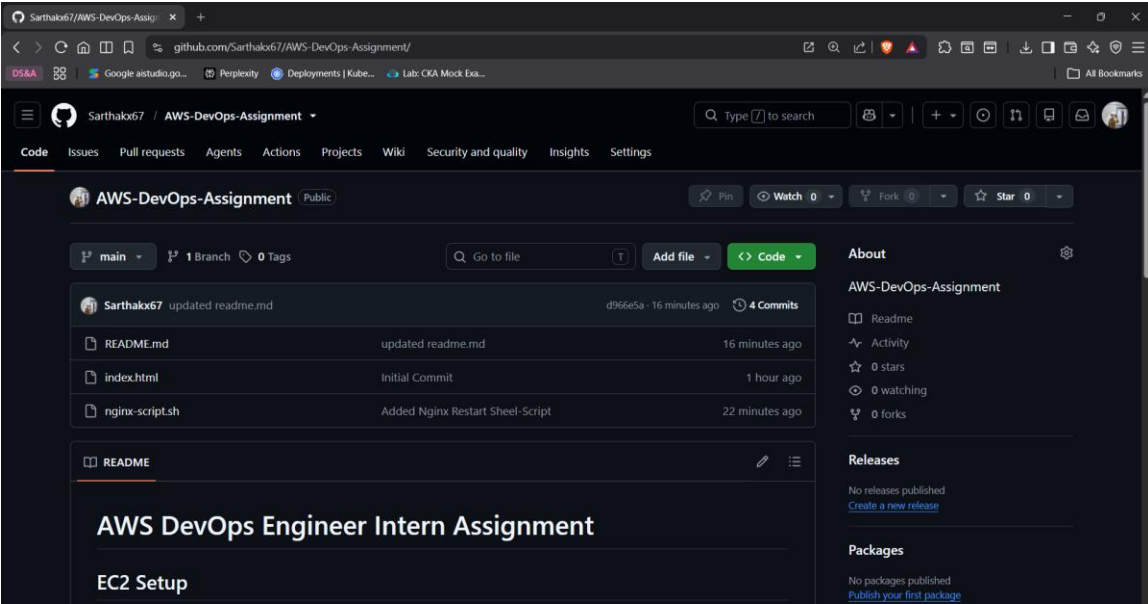
For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@ip-172-31-5-142:~$
```

Task 4: Git & GitHub

Repository Link:

<https://github.com/Sarthakx67/AWS-DevOps-Assignment/>



Time Taken

Approximate total time: 30 min + Documentation

Conclusion

deployed an Ubuntu EC2 instance, configured Nginx, hosted a custom HTML webpage, and completed bonus tasks using Docker and a shell script. This assignment provided hands-on experience with AWS, Linux, Docker, and basic DevOps practices.